

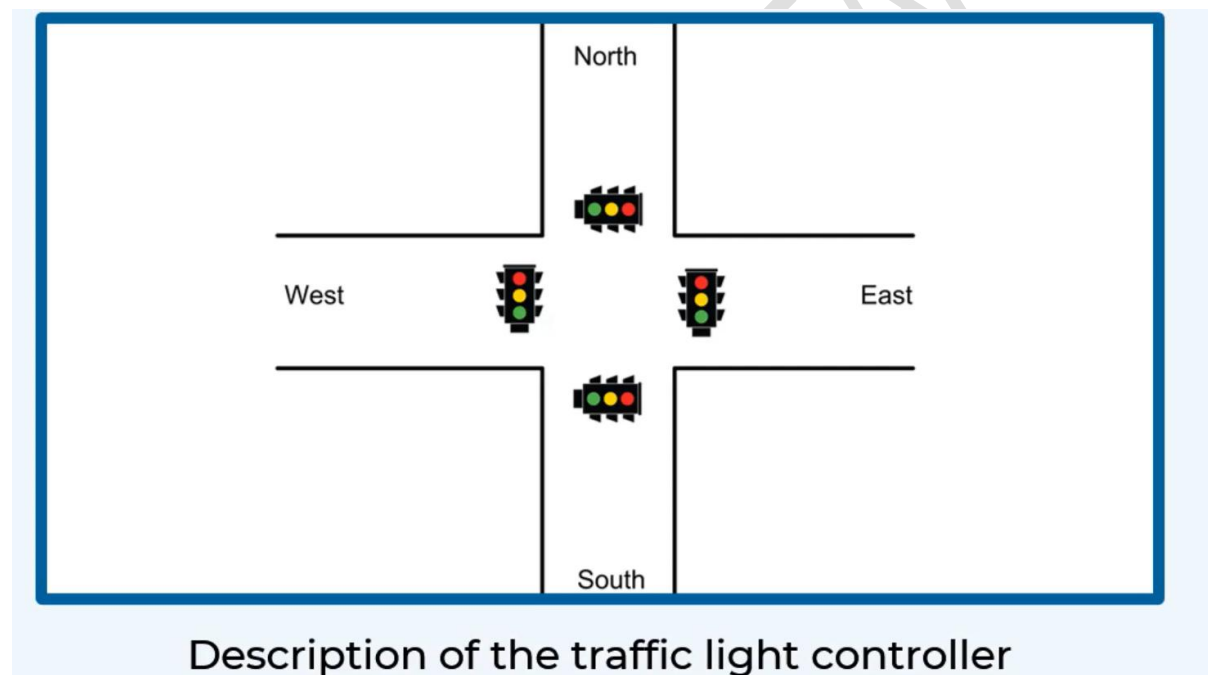
Project – Traffic Light Controller

Problem Statement :

- A traffic light is to be installed on the intersection of 4 ways.
- The directions are assumed as east, west, north and south.
- Only one direction of traffic is allowed to move at a time
- Red, Yellow, and Green signals control the traffic of each direction.

4-way traffic signal controller design

- Every light is set to green for 30 seconds.
- Then, it is set to yellow for 5 seconds.
- Repeat the same process for other directions.
- Other lights remain red during these moment.



Verilog design:

- For each color, we can assign numbers for easy implementation.
- Each direction can also be coded into the numbers.
- Clock signal is used to synchronize all the lights.
- Reset signal is used to reset the controller.

Verilog Code:

```
module Traffic_sys(north_light,west_light,south_light,east_light,clk,reset);

    output reg [2:0] north_light,west_light,south_light,east_light;

    input    clk;
    input    reset;

    reg [2:0] state;

    parameter [2:0] north=3'b000, north_yellow=3'b001, west=3'b010, west_yellow=3'b011,
    east=3'b100, east_yellow=3'b101, south=3'b110, south_yellow=3'b111;

    reg [3:0] count;

    always @(posedge clk, posedge reset)
    begin
        if (reset)
            begin
                state=north;
                count =3'b0000;
            end
        else
            begin
                case (state)
                    north :
                        begin
                            if (count==4'b1111)
                                begin
```

```
        count=4'b0000;  
        state=north_yellow;  
    end  
else  
    begin  
        count=count+4'b0001;  
        state=north;  
    end  
end
```

north_yellow :

```
begin  
    if (count==4'b0011)  
        begin  
            count=4'b0000;  
            state=west;  
        end  
    else  
        begin  
            count=count+4'b0001;  
            state=north_yellow;  
        end  
end
```

south :

```
begin  
    if (count==4'b1111)  
        begin
```

```
        count=4'b0000;  
        state=south_yellow;  
    end  
else  
    begin  
        count=count+4'b0001;  
        state=south;  
    end  
end
```

south_yellow :

```
begin  
    if (count==4'b0011)  
        begin  
            count=4'b0000;  
            state=east;  
        end  
    else  
        begin  
            count=count+4'b0001;  
            state=south_yellow;  
        end  
    end
```

east :

```
begin  
    if (count==4'b1111)  
        begin
```

```
        count=4'b0000;  
        state=east_yellow;  
    end  
else  
    begin  
        count=count+4'b0001;  
        state=east;  
    end  
end
```

east_yellow :

```
begin  
    if (count==4'b0011)  
        begin  
            count=4'b0000;  
            state=north;  
        end  
    else  
        begin  
            count=count+4'b0001;  
            state=east_yellow;  
        end  
end
```

west :

```
begin  
    if (count==4'b1111)  
        begin
```

```
        state=west_yellow;
        count=4'b0000;
    end
else
    begin
        count=count+4'b0001;
        state=west;
    end
end

west_yellow :
begin
    if (count==4'b0011)
        begin
            state=south;
            count=3'b0000;
        end
    else
        begin
            count=count+4'b0001;
            state=west_yellow;
        end
    end

endcase // case (state)

end // always @ (state)

end
```

```
always @(state)
```

```
begin
```

```
case (state)
```

```
  north:
```

```
    begin
```

```
      north_light = 3'b001;
```

```
      south_light = 3'b100;
```

```
      east_light = 3'b100;
```

```
      west_light = 3'b100;
```

```
    end
```

```
  north_yellow :
```

```
    begin
```

```
      north_light = 3'b010;
```

```
      south_light = 3'b100;
```

```
      east_light = 3'b100;
```

```
      west_light = 3'b100;
```

```
    end
```

```
  south :
```

```
    begin
```

```
      north_light = 3'b100;
```

```
      south_light = 3'b001;
```

```
      east_light = 3'b100;
```

```
      west_light = 3'b100;
```

```
    end
```

```
  south_yellow :
```

```
begin
    north_light = 3'b100;
    south_light = 3'b010;
    east_light = 3'b100;
    west_light = 3'b100;
end
```

west :

```
begin
    north_light = 3'b100;
    south_light = 3'b100;
    east_light = 3'b100;
    west_light = 3'b001;
end
```

west_yellow :

```
begin
    north_light = 3'b100;
    south_light = 3'b100;
    east_light = 3'b100;
    west_light = 3'b010;
end
```

east :

```
begin
    north_light = 3'b100;
    south_light = 3'b100;
    east_light = 3'b001;
```

```

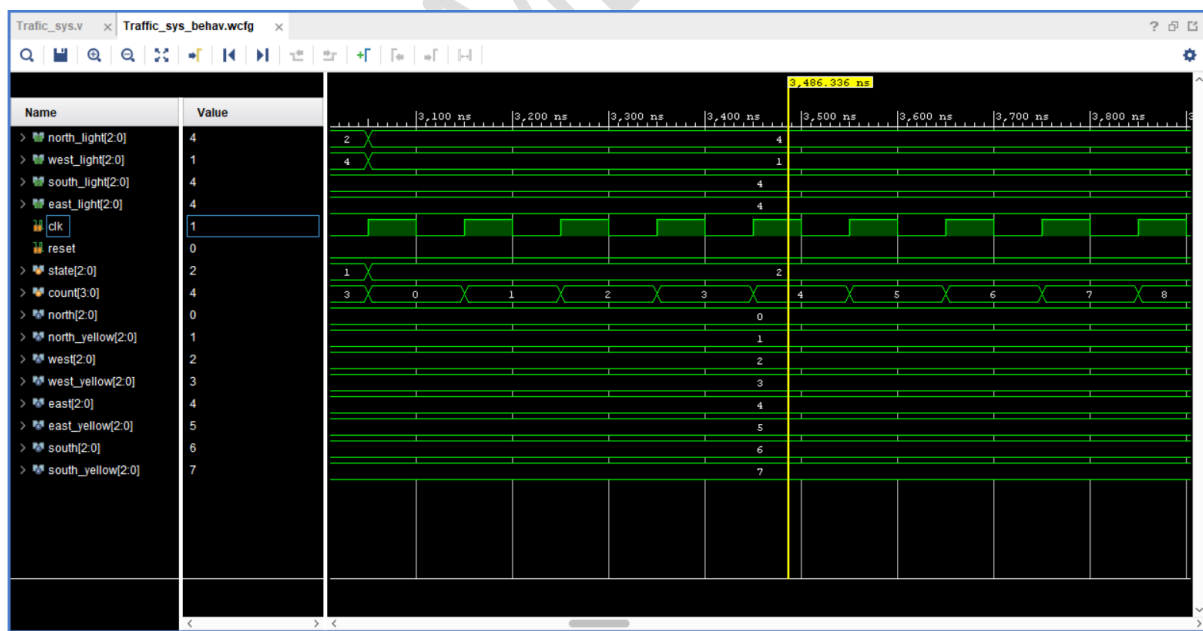
        west_light = 3'b100;
    end

    east_yellow :
        begin
            north_light = 3'b100;
            south_light = 3'b100;
            east_light = 3'b010;
            west_light = 3'b100;
        end
    endcase

end
endmodule

```

Simulation Waveform :



Tools used and Language Used:

Xilinx Vivado and Verilog.